

LLM Tutorial

Applying LLMs to Synthetic Healthcare Data for Disease Prediction and Clinical Note Generation



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Introduction

Background: Large Language Models (LLMs) are being explored for automating healthcare tasks such as summarization, prediction, and patient communication.

Objective:

- Apply multiple LLM reasoning & prompt engineering strategies to healthcare data.
- Evaluate performance across structured-to-text, classification, and simplification tasks.

Data Privacy:

Used **Synthea synthetic patient data** (100 patients, no real PHI).

Data used

Source: Synthea 100-patient CSV bundle.

Key features used:

- Demographics: age, sex
- Diagnoses (ICD-10 codes)
- Medications
- Labs: A1C, LDL, BMI, Systolic BP, Diastolic BP

Target label: Diabetes (yes/no) based on ICD-10 codes E10/E11 or diagnosis text.

Sample size:

- Balanced subset for classification: 80 records
- Smaller subsets for generation: 20–40 records per method for speed.

Data Preview 1

```
print("patients:", patients.shape)
print("conditions:", conditions.shape)
print("observations:", observations.shape)
print("medications:", medications.shape)

assert not patients.empty, "patients.csv not found or empty. Fix DATA_DIR and rerun."
```

```
patients: (111, 28)
conditions: (4140, 7)
observations: (88156, 9)
medications: (4926, 13)
```

```
patients.columns
```

```
Index(['Id', 'BIRTHDATE', 'DEATHDATE', 'SSN', 'DRIVERS', 'PASSPORT', 'PREFIX',
      'FIRST', 'MIDDLE', 'LAST', 'SUFFIX', 'MAIDEN', 'MARITAL', 'RACE',
      'ETHNICITY', 'GENDER', 'BIRTHPLACE', 'ADDRESS', 'CITY', 'STATE',
      'COUNTY', 'FIPS', 'ZIP', 'LAT', 'LON', 'HEALTHCARE_EXPENSES',
      'HEALTHCARE_COVERAGE', 'INCOME'],
      dtype='object')
```

Data Preview 2

[8] observations.head()

	DATE	PATIENT	ENCOUNTER	CATEGORY	CODE	DESCRIPTION	VALUE	UNITS	TYPE
0	2017-12-22T02:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	1f9f55cd-ae88-b9ab-15e1-927e73848ca3	vital-signs	8310-5	Body temperature	37.8	Cel	numeric
1	2018-04-04T06:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	e2113f4b-89c7-a231-d3ae-a6a6b4f41566	vital-signs	8302-2	Body Height	185.6	cm	numeric
2	2018-04-04T06:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	e2113f4b-89c7-a231-d3ae-a6a6b4f41566	vital-signs	72514-3	Pain severity - 0-10 verbal numeric rating [Sc...	0.0	[score]	numeric
3	2018-04-04T06:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	e2113f4b-89c7-a231-d3ae-a6a6b4f41566	vital-signs	29463-7	Body Weight	104.5	kg	numeric
4	2018-04-04T06:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	e2113f4b-89c7-a231-d3ae-a6a6b4f41566	vital-signs	39156-5	Body mass index (BMI) [Ratio]	30.3	kg/m2	numeric

medications.head()

	START	STOP	PATIENT	PAYER	ENCOUNTER	CODE	DESCRIPTION	BASE_COST	PAYER_COVERAGE	DISPENSES	TOTALCOST	REASONCODE	REASONDESCRIPTION
0	2016-04-11T11:48:33Z	2016-04-26T11:48:33Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	8fa6c185-e44e-3e34-8bd8-39be8694f4ce	87741be8-2223-14c5-375d-976414fcd9aa	313782	Acetaminophen 325 MG Oral Tablet	119.52	0.0	1	119.52	NaN	NaN
1	2018-02-24T12:17:26Z	2018-03-21T12:17:26Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	8fa6c185-e44e-3e34-8bd8-39be8694f4ce	0d6b6a0a-8f84-c860-ba6f-b241d14f55bf	849574	Naproxen sodium 220 MG Oral Tablet	91.72	0.0	1	91.72	NaN	NaN
2	2024-10-05T22:04:53Z	2024-10-23T22:04:53Z	8c8e1c9a-b310-43c6-33a7-ad11bad21c40	8fa6c185-e44e-3e34-8bd8-39be8694f4ce	1cd44e1e-0c9f-ab84-f187-11f0418849f4	313782	Acetaminophen 325 MG Oral Tablet	45.86	0.0	1	45.86	10509002.0	Acute bronchitis (disorder)
3	2017-06-19T01:38:45Z	2017-07-01T01:38:45Z	edc17058-55fb-08c7-12df-ece93a402e50	8fa6c185-e44e-3e34-8bd8-39be8694f4ce	3f9da7ae-4222-4094-dc1b-e30b9189d650	562251	Amoxicillin 250 MG / Clavulanate 125 MG Oral T...	252.65	0.0	1	252.65	444814009.0	Viral sinusitis (disorder)
4	2020-07-12T09:38:45Z	2020-07-25T09:38:45Z	edc17058-55fb-08c7-12df-ece93a402e50	8fa6c185-e44e-3e34-8bd8-39be8694f4ce	329ee98f-4c76-5422-653d-03fc8eb0bcf9	1043400	Acetaminophen 21.7 MG/ML / Dextromethorphan Hy...	61.15	0.0	1	61.15	10509002.0	Acute bronchitis (disorder)

Task S

- **Table to Text:** Generate a concise clinical snapshot from structured data.
- **Disease Classification:** Predict diabetes risk (yes/no/unsure).
- **Simplification:** Rewrite clinical note for a specific audience (older adult, grade 6).

Prompting Methods Used

Zero-Shot: Task instructions only, no examples.

Few-Shot: Task instructions + 2–3 labeled examples to guide format/style.

Chain-of-Thought (CoT):

- Explicitly instruct model to **list reasoning steps first** before giving final answer.
- Used in: **reasoned** prompts for classification & table→text.

Tree-of-Thought (ToT):

- Generate **two candidates** using different prompting strategies.
- Ask a separate "judge" prompt to pick the best output.
- Used in: **method="tot"** for table→text.

Self-Consistency (Sc-3)

- Run CoT + few-shot classification **3 times**.
- Use majority vote for final label.
- Bonus point: combines reasoning + ICL + decision layer.

Zero-Shot Prompt

Zero-Shot: Task instructions only, no examples. (Refer to code for more such prompts)

Task Example: Convert structured data to Clinical snapshot (table -> text)

Prompt

System:

You write concise clinician-facing patient snapshots. Do not invent fields.

User:

You'll receive a JSON record delimited by <record>. Use only its content.

Rules:

- Cover: Problems, Meds, Labs (flag highs/lows), Risks, Next Steps.
- Quote numeric values with units if present.
- If info is missing, say "not recorded".
- Keep to 120–150 words. No headings. No lists.

<record>

{{ record_json }}

</record>""

Few-Shot Prompt

Few-Shot: Task instructions + 2–3 labeled examples to guide format/style. (Refer to code for more such prompts)

Task Example:
Classification of Diabetic patient.
Predict Diabetes risk (yes/no/unsure)

Prompt

System:

You are a careful clinical classifier. If uncertain, return "unsure".

User:

Classify the record between {"yes","no","unsure"} for diabetes, and estimate risk {"low","medium","high"}.

Decision rules:

- Existing diabetes diagnosis → "yes".
- $\text{HbA1c} \geq 6.5\% \rightarrow \text{"yes"}$ (cite value).
- $\text{HbA1c } 6.0\text{--}6.4\%$ with other diabetes signals → "unsure".
- No signal → "no".

Output JSON ONLY:

```
{"label":"yes|no|unsure","risk":"low|medium|high","rationale":"≤30 words citing fields"}
```

Example In:

```
{"age":54,"sex":"F","diagnoses":["hypertension"],"labs":{"A1C_%":7.1}}
```

Example Out:

```
{"label":"yes","risk":"medium","rationale":"A1C 7.1% ≥ 6.5% threshold."}
```

Now classify:

```
<record>
```

```
{{ record_json }}
```

```
</record>
```

Chain-of-Thought

Chain-of-thought: Explicitly instruct model to **list reasoning steps first** before giving final answer.
(Refer to code for more such prompts)

Task: Classification of Diabetic patient. Predict Diabetes risk (yes/no/unsure)

Prompt

System:

You are a careful clinical classifier. If uncertain, return "unsure".

User:

First, **list the signals** that support or refute diabetes based on the record. Then output the final JSON ONLY as last line:

```
{"label":"yes|no|unsure","risk":"low|medium|high","rationale":"≤30 words citing fields"}.
```

Record:

```
<record>
```

```
{{ record_json }}
```

```
</record>
```

Tree-of-Thought

Tree-of-Thought: Generate **two candidates** using different prompting strategies. Ask a separate "judge" prompt to pick the best output. (Refer to code for more such prompts)

Task: Convert structured data to Clinical snapshot (table -> text)

```
elif method == "tot":
    p1 = render("table2text_reasoned", record_json=json.dumps(rec, ensure_ascii=False))
    p2 = render("table2text_fewshot", record_json=json.dumps(rec, ensure_ascii=False))
    c1 = gen_text(p1, temperature); c2 = gen_text(p2, temperature)
    judge = f"You are a clinician. Pick the better snapshot (A or B) based on factuality, correct units, completeness. Respond 'A' or 'B' only.\nA) {c1}\nB) {c2}\n"
    pick = gen_text(judge, temperature=1)
    out = c1 if pick.strip().upper().startswith("A") else c2
    rows.append({"id": r["patient_id"], "task": "table2text", "method": method, "output": out})
    continue
else:
```

Self-Consistency (SC-3)

Self Consistency: Run CoT + few-shot classification 3 times. Uses the majority vote for final label. Combines reasoning + ICL + decision layer

Task: Classification of Diabetic patient. Predict Diabetes risk (yes/no/unsure)

```
elif method == "reasoned_fewshot":
    base = PROMPTS["disease_cls_reasoned"] + \
        '\n\nExample In:\n{"age":54,"sex":"F","diagnoses":["hypertension"],"labs":{"A1C_%":7.1}}\n' + \
        'Example Out:\n{"label":"yes","risk":"medium","rationale":"A1C 7.1% ≥ 6.5% threshold."}\n'
    prompt = Environment(loader=DictLoader({"tmp": base})).get_template("tmp").render(
        record_json=json.dumps(rec, ensure_ascii=False)
    )
    outs = [safe_json(gen_text(prompt, temperature)) for _ in range(samples)]
```

Model Used and Experiment

Gen backend: OPENAI

Model Used: Gpt-5-mini

Temperature Used: 1

Experiments:

Task 1: Simplification of medical notes for the younger adults.

- N = 40 for zero-shot prompt

Task 2: Diabetes risk prediction (BONUS POINT EFFORT)

- N = 60 for Self consistency -> Chain-of-thought + 3 few shot examples
- Incorporated Decision Layer

Task 3: Clinical snapshot generation from the structured data.

- N = 20 each for zero-shot, few-shot, chain-of-thought and tree-of-thought

Evaluation and Results

Classification task: Diabetes risk prediction

Experiment: N = 60 for Self consistency -> Chain-of-thought + 3 few shot examples | Incorporated Decision Layer

Metrics:

1. **Coverage:** % of predictions that are definite yes / no (not “unsure”) -> **83.3%**
2. **Accuracy:** proportion of correct predictions made by a model out of the total number of predictions. -> **58%**
3. **Precision:** Accuracy of positive prediction made by the model. -> **48.8%**
4. **Recall:** Proportion of actual positives that are correctly classified as positive by the model. -> **100%**
5. **F1-score:** Metric that provides a balanced measure of a model's precision and recall. -> **65.6%**

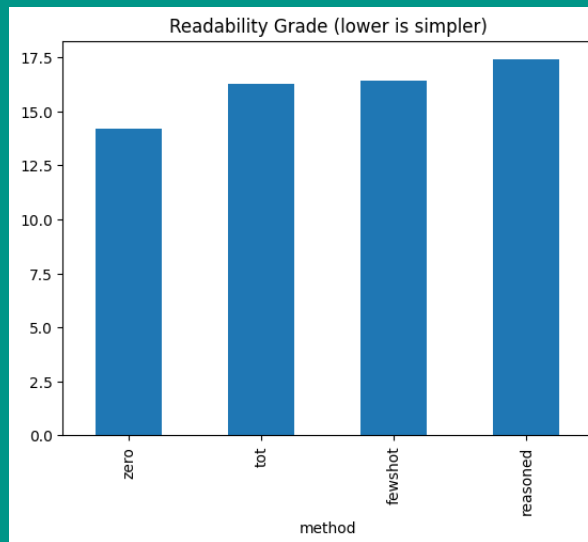
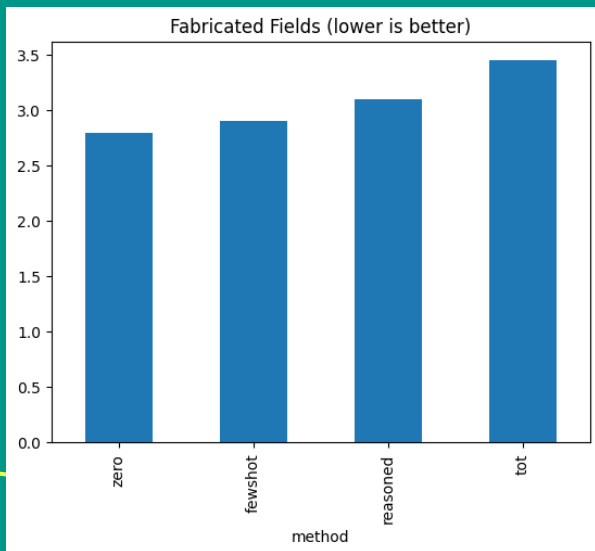
Confusion Matrix	Predicted: Yes	Predicted: No
Actual: Yes	20	0
Actual: No	21	9

Evaluation and Results

Table to Text: Convert Structured data to Clinical Snapshots

Metrics:

1. **Fabricated Fields:** Count of invented values not in input table.
2. **Unit Sanity:** Check if key metrics include correct units (e.g., “mg/dL” for LDL).
3. **Readability:** Flesch-Kincaid grade level (lower = easier to read).



Evaluation and Results

Simplification: Convert Health explanations into the plain language

Output:

What it means - You are an adult woman with several health problems. - You have long-lasting sinus infections (chronic sinusitis), anemia, and prediabetes. - A1C 5.8% and BMI 27.6 show your sugar and weight are a bit high. - Your blood pressure is 120/91 mmHg. - The plan is to reduce risks and follow up with your doctor. What to do - Take medicines exactly as prescribed. - Keep follow-up visits so your doctor can watch your health. When to seek help - Call your doctor if you get worse, have trouble breathing, or have new or bad side effects from medicines.

Metrics:

1. **Readability** improvement (lower FK grade). -> **71.6%**
2. **Jargon Rate**: % of domain-specific terms retained. -> **0.3%**

Reflections

Prompting matters: Few-shot improves style, CoT improves reasoning, ToT reduces factual errors.

Bonus approach: CoT + ICL + Self-Consistency gave robust classification.

Synthetic data is safe, but results may differ with real EHR data.

Future work:

- Multi-condition prediction
- RAG with real clinical guidelines
- Fine-tuned models for healthcare domain



Thankyou!